



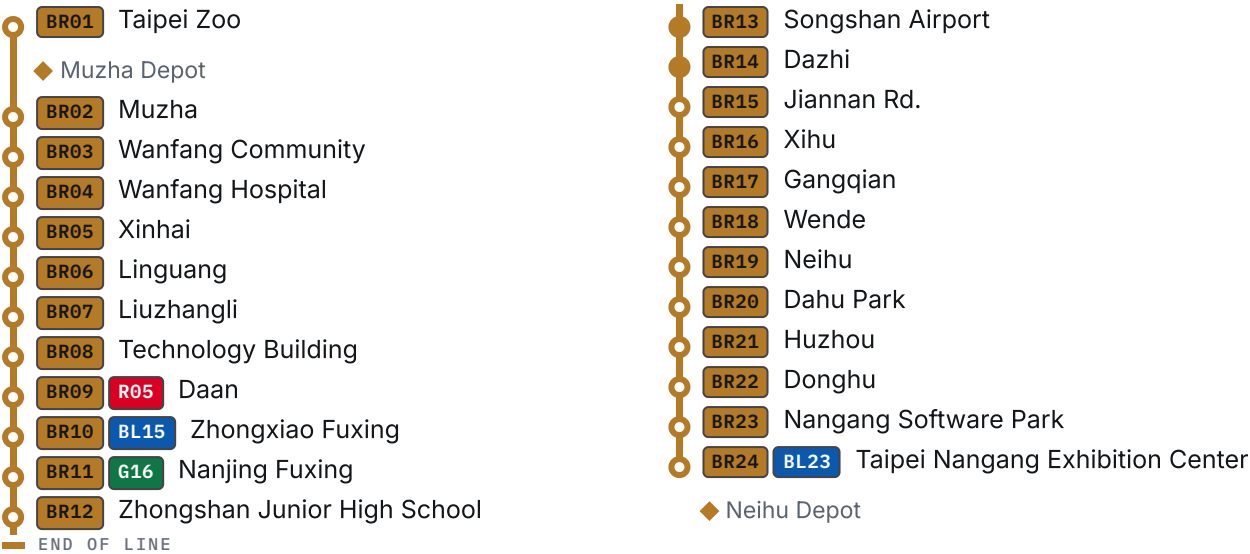
A VAL256 crosses the Keelung River into Neihu.

玄史生 [https://commons.wikimedia.org/wiki/File:VAL_256_train_across_Keelung_River_on_Taipei_Metro_Neihu_20141030.JPG] · CC0 [<http://creativecommons.org/publicdomain/zero/1.0/deed.en>] · [WIKIMEDIA COMMONS](#)

Wenhu Line

Taipei's driverless brown line — no cab, a front window you can stand at, and a rubber-tyred guideway that climbs hills the rest of the network cannot.

THE LINE



○ ELEVATED ● UNDERGROUND

25.17 km^[1]

OPERATING LENGTH, BR01 TO BR24

Also published as 25.7 km (two construction projects added together), 25.04 km (this site's own measurement) and 26.42 km (the whole alignment, depot leads included). Four figures, four different things — the section below says which is which.

BR WENHU LINE 文湖線			
LINE CODE BR	OPERATOR Taipei Rapid Transit Corporation	TERMINI BR01 — BR24	STATIONS 24 ^[2]
ELEVATED 22 ^[2]	UNDERGROUND BR13, BR14 ^[6]	OPENED 28 March 1996 ^[2]	LAST EXTENSION 4 July 2009 ^[2]
END TO END 45 min ^[7]	GUIDEWAY Rubber-tyred, side-guided ^[14]	ELECTRIFICATION 750 V DC, off the guide bars ^[14]	SIGNALLING CITYFLO 650, Bombardier (now Alstom) ^[8]
AUTOMATION GoA4, driverless ^[8]	DEPOTS Muzha, Neihu		

There is no cab. Stand at the front of the leading car and the guideway comes straight at you — through the hills above Muzha [\[/rail/stations/br02/\]](/rail/stations/br02/), across the rooftops of eastern Taipei, over the Keelung River into Neihu [\[/rail/stations/br19/\]](/rail/stations/br19/). It is the best free view in the city, and it exists because nobody needs to sit where you are standing.

The Wenhua Line was **Taiwan's first metro line** ^[15], and it remains the only line in the Taipei network running on rubber tyres rather than steel wheels. That choice is what lets it climb through Wenshan's hills; the precise gradients it clears are a figure this site has not yet sourced, and the design criteria that would settle them are still to be found.

History

The name is a contraction: **Wenshan** and **Neihu**, the districts at either end. The two halves opened thirteen years apart, and they are the two halves of one of the strangest procurement stories in modern transit.

STAGE	SECTION	LENGTH	OPENED
Muzha Line	BR01 — BR12	10.9 km ^[2]	28 March 1996 ^[2]
Neihu extension	BR13 — BR24	14.8 km ^[2]	4 July 2009 ^[2]

The alignment was redrawn by protest before a metre of it was built

The route that opened in 1996 is not the route that was approved. A 13 km scheme from Taipei Zoo [\[/rail/stations/br01/\]](/rail/stations/br01/) to Liuzhangli [\[/rail/stations/br07/\]](/rail/stations/br07/) had already been settled when **residents in Muzha objected, and on 4 May the northern section was redirected onto Fuxing South and North Roads** — where it runs today ^[9].

That is the earliest instance of a pattern this site records twice more: the Songshan–Xindian Line [\[/rail/lines/songshan-xindian-line/\]](/rail/lines/songshan-xindian-line/) redesigned from part-elevated to fully underground in 1987 on street-width and visual-impact grounds, and the Neihu extension held up for years over the same argument. **The objection came first, before the line existed, and it worked.** What exactly was moved, and which proposed stations went with it, is not established here and is worth a proper look.

The original section was bought in 1988 as a turnkey system from France's Matra ^[9]. What followed — **two fires during testing in 1993**, on 5 May and 24 September, the second recorded as exactly that, 第二次火燒車事件, a second vehicle-fire incident ^[3]; an opening four and a quarter years late; the builder's withdrawal from maintaining its own system; and a lawsuit that ran until the Supreme Court ordered the city to pay roughly NT\$1.64 billion in 2005 ^[9] ^[10] — is told in full on its own page: **The Matra dispute** [\[/rail/history/matra-dispute/\]](/rail/history/matra-dispute/). The short version is that Taipei learned, expensively, how not to buy a railway, and bought the second half differently.

When the Neihu extension opened on 4 July 2009, the two sections began running as one through line under the extension's new CITYFLO 650 signalling ^[2] ^[8]. The original VAL256 fleet was withdrawn that same day for conversion to the new system and did not carry passengers again until 26 December 2010 ^[11] — eighteen months in which the 1996 railway ran entirely on the 2009 trains. See VAL256 [\[/rail/rolling-stock/val256/\]](/rail/rolling-stock/val256/).

Route

Elevated for almost its entire length. From **BR01** the line climbs north through Wenshan District, crosses central Taipei on an east–west alignment, then turns north-east across the Keelung River into Neihu.

Two of the twenty-four stations are underground: **BR13 Songshan Airport** [/rail/stations/br13/] and **BR14 Dazhi** [/rail/stations/br14/], and both are now settled by DORTS itself rather than inferred.

The builder's project record gives the Neihu section as 14.8 km with 「地下段3.9公里，2座地下站」 — 3.9 km underground, two underground stations^[2] — which narrows the pair to **BR13** — **BR24** without naming them. Its station architecture page then names one of the two and counts the rest: 「除大直站為地下車站外，其餘共有十座高架車站」, apart from Dazhi being underground, the remaining ten are elevated^[6]. Ten elevated plus Dazhi is eleven of the section's twelve stations, and the project record says there are exactly two underground^[2]. The twelfth station is Songshan Airport, so Songshan Airport is the second. Two primary documents that never mention each other, and the identification falls out of the arithmetic between them. zh.wikipedia names the same pair outright^[8]; it is no longer what the claim rests on.

The platform depths are still not established here and are not asserted.

Every platform on the line has full-height platform screen doors: Faiveley equipment from **BR01** to **BR12**, ST Electronics from **BR13** to **BR24**^[13].

How many ways out

DORTS's station table counts the street exits at every station, and the distribution is lopsided in a way that says something about how the line was built^[2].

Eight of the twenty-four stations have exactly one exit^[2], and seven of those eight are consecutive: **BR02** Muzha through **BR08** Technology Building, an unbroken run from Wenshan into Daan [/rail/stations/br09/] where every station has one way in and one way out. **BR12** Zhongshan Junior High School [/rail/stations/br12/] is the eighth^[2].

Twenty of the twenty-four have three exits or fewer. The whole line has sixty-three between its twenty-four stations, and a quarter of those are at two stations: **BR11** Nanjing Fuxing [/rail/stations/br11/] and **BR24** Nangang Exhibition Center have eight each. **BR09** Daan has six, **BR10** Zhongxiao Fuxing [/rail/stations/br10/] five. All four are interchanges^[2].

That is not a trivia column. A single-exit station is one lift, one stair, one escalator bank and one crowd; when it closes, the station closes. It is a predictable consequence of building a medium-capacity railway on a single row of columns down the middle of an existing road — the same constraint DORTS gives for why these stations carry no decoration, which is that a single-column support system limits what you can hang on it^[6]. The exits are the same economy, at street level.

Every station page carries its own count, from the same table^[2].

BR06 **Linguang** [/rail/stations/br06/] **is the exception that proves the constraint was real.** It has two exits and it is the first station on the Taipei Metro [/rail/operators/trtc/] that opened with both of them already built^[3] — everywhere else, a second exit was something added later or not at all. The same station has **two ticket machines, the fewest anywhere on the network**^[3].

Operations

Trains run without on-board drivers under automatic train control, at GoA4 — unattended train operation, the highest grade^[8].

Weekday headways are about 2–4 minutes at peak, 4–10 off-peak, and about 12 minutes after 23:00; weekends and holidays run the off-peak pattern all day^[7]. **End to end takes about 45 minutes**, **BR01** to **BR24** or the reverse^[7]. zh.wikipedia adds that the headway can be compressed to 80 seconds^[8]; that is a signalling capability rather than a timetable, and no primary source for it was found.

How long the line is

Four figures circulate, and they are not in conflict — they measure four different things. **This site publishes 25.17 km, the operating length**, and says what each of the others counts.

FIGURE	WHAT IT MEASURES	SOURCE
25.17 km	Operating length — the railway that carries passengers, BR01 to BR24	TDX <code>CumulativeDistance</code> against the last station of route BR-1 ^[1]
25.7 km	The two construction projects added together	10.9 km + 14.8 km, both DORTS's own figures ^[2]
25.04 km	The same railway, measured by this site along MOTC's surveyed geometry from the first station to the last	the map below
26.42 km	The whole published alignment, including 635 m of depot lead beyond BR01 and 744 m of tail track beyond BR24	the same geometry, untrimmed

The last row is the one this site got wrong: it printed the untrimmed figure as the route length for four builds, because measuring everything MOTC draws counts depot lead and tail track that no passenger travels on^[1]. Trimmed to the termini, the same geometry lands within 130 m of the operator's own figure^[1] — and that agreement is the check working, which is why both are still printed side by side on the network page [/rail/network/].

The 25.7 km that appears as 路線長度 on zh.wikipedia^[8] is the **sum of the two published section lengths**: DORTS records 木柵線 約10.9公里 and 內湖線 約14.8公里, and $10.9 + 14.8 = 25.7$ exactly^[2]. Those are project lengths — the extent of what was built, including non-revenue track at each end — which is why the total exceeds the 25.17 km of railway that carries passengers^[1].

Rolling stock

Two fleets share the line, both running as four cars made up of two coupled pairs:

- VAL256 [/rail/rolling-stock/val256/] — 51 pairs, the original Muzha Line fleet^[11]
- Innovia APM 256 (C370) [/rail/rolling-stock/innovia-apm-256-c370/] — 101 pairs, delivered for the Neihu extension^[12]

The fleet totals do not divide, and this site cannot say why

A four-car train on this line is two coupled pairs. DORTS's procurement record gives **51 pairs of VAL256 and 101 pairs of Innovia**^[4]. The renewal reporting counts **25 Matra trains and 51 Bombardier trains**, 76 in total^[5].

Those do not reconcile^[4] ^[5]. 25 four-car trains need 50 pairs, not 51; 51 trains need 102 pairs, not 101. The totals agree — 152 pairs, 304 cars, 76 trains — but **neither fleet on its own divides into whole trains**.

Three readings, and nothing found settles between them:

- **A spare pair in each fleet.** 25 Matra trains plus one pair out of service, 50 Bombardier trains plus one — which makes the "76 trains" figure a count of the two fleets' nominal sizes rather than of trains in traffic.
- **Formations are not fixed.** zh.wikipedia describes the VAL256 as 「非固定編組」, running as four consecutively-numbered cars rather than as a permanent set^[5]. If pairs are re-formed in maintenance, the two counts are simply two different things measured on two different days^[5], and there is no discrepancy to explain.
- **One train is mixed.** Moving a pair across the boundary makes both counts work — and would mean exactly one train on this line runs one fleet's pair coupled to the other's.

The third would make this page wrong, because it describes the fleets as separate. It is also the only one of the three that requires something no source found here reports^[5]. It is recorded here because leaving it out would be picking an answer quietly.

They are not the same vehicle and not the same design lineage — the second was built by a competitor to fit the first's guideway, which is the Matra story's last chapter — but the interface they share is exact: the same running-surface spacing, the same side guide bars, the same 750 V taken off them^[12] ^[14]. TRTC has announced a programme to replace all 76 trains by 2041 — see the research notes, and treat the figures as provisional.

Depots

Muzha Depot `[/rail/depots/muzha-depot/]` sits at the southern end beside `BR01` and opened with the line in 1996^[2]. Neihu Depot `[/rail/depots/neihu-depot/]` was built for the extension and came into use in 2009^[2]. Despite its name it is in Nangang District, and it joins the running line at `BR24`, at the far end of the line^[2].

Sources

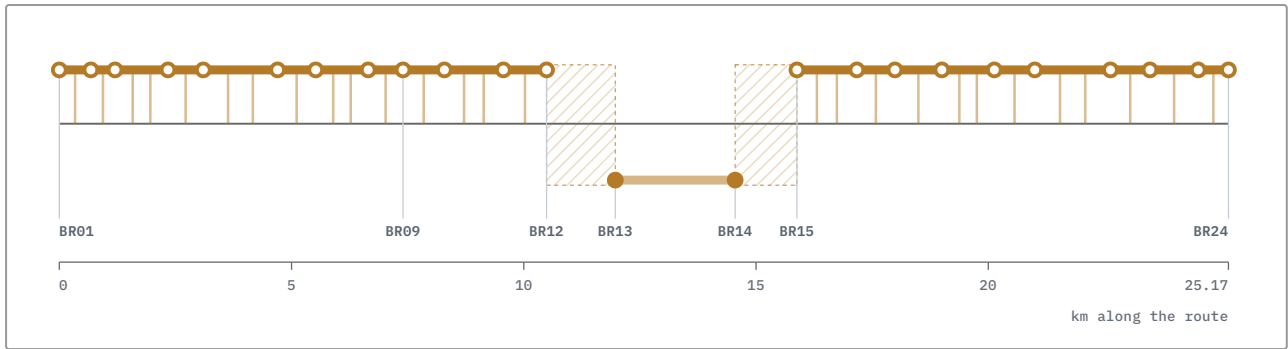
Station names, codes, running order, coordinates and the end-to-end journey time come from **Taiwan MOTC's TDX platform**, operator TRTC — government open data, retrieved 5 August 2026. The strip map is generated from it directly. Everything else is cited inline and listed below.

TDX does not publish station structure, headways, or anything about rolling stock and depots; those come from TRTC's and DORTS's own publications where they exist, and are marked secondary where they do not.

Corrections

This page's errors are corrected in place and recorded here, most recent first.

- **The strip map drew Neihu Depot at the wrong end of the line for four builds.** This page previously said the depot joined the running line near `BR19` Neihu — an unsourced inference from the shared name. It actually connects at `BR24`, in Nangang District^[2], and the map marker moved with the fact. `npm run facts` now cross-checks the marker against the depot page's own stated junction.
- **Two engineering superlatives were removed rather than corrected.** This page used to say the gradients through Wenshan were steeper than conventional heavy metro could manage, and that the elevated curves would be *impossible* on a steel-wheel alignment. Both are probably true and neither was sourced, and "impossible" was doing a great deal of unattributed work. They can return when DORTS's design criteria are found.
- **The 25.7 km figure was described as unaccounted for.** An earlier version of this page could not say where the widely-circulated 25.7 km came from. It is simply the sum of DORTS's two project lengths, as the length section above now shows.

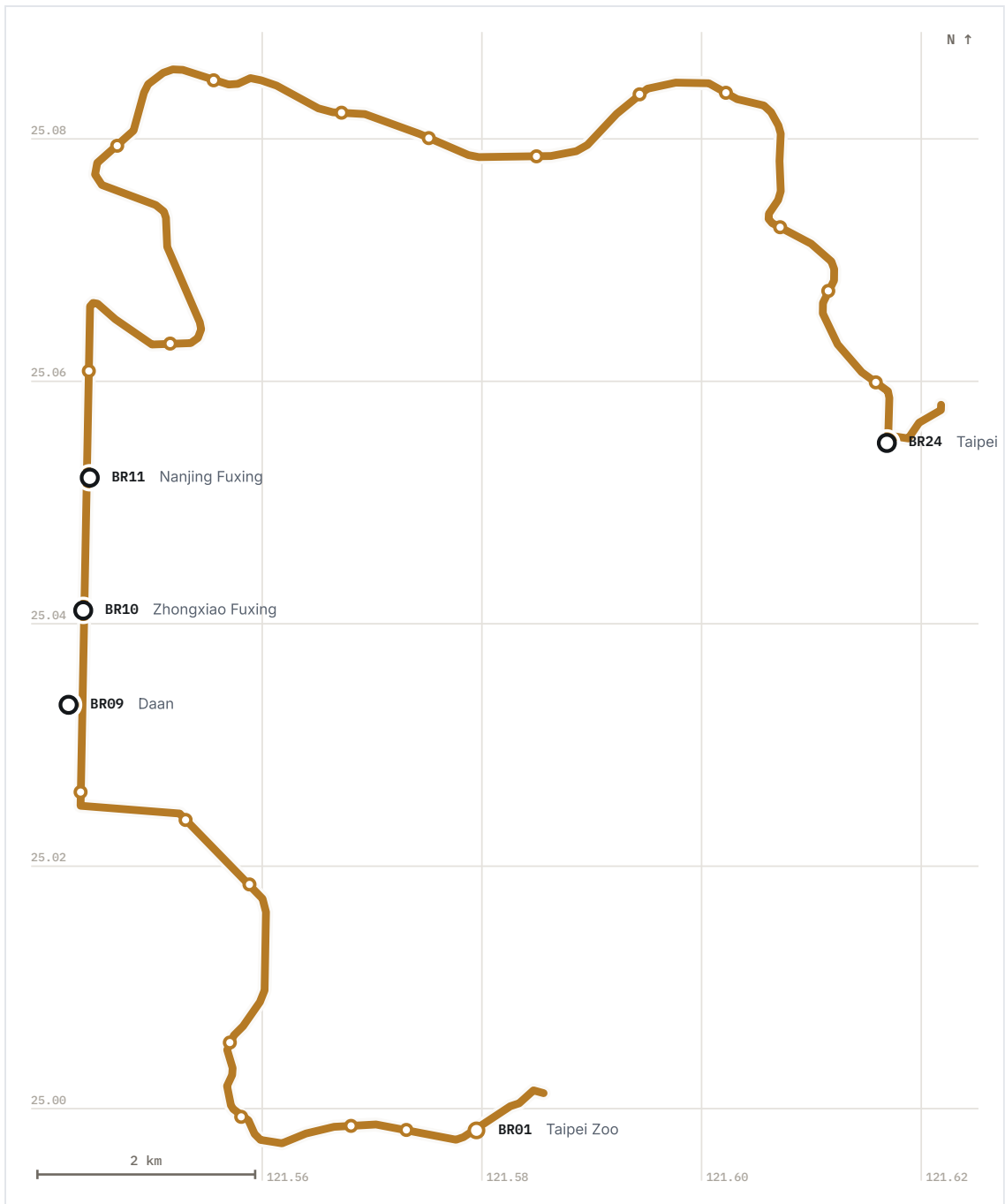


elevated — deck on piers
 underground

structure changes somewhere in here; no source says where

The Wenhua Line in section. It runs on viaduct for all but 3.9 km of its length, dives under Songshan Airport and the Keelung River, and comes back up beside the south portal of the Ziqiang road tunnel at Dazhi. **3.9 km of the line runs underground.** 2.58 km of that is between the two underground stations, which is fixed by the operator's own chainage. The remaining 1.32 km is ramp at the two portals, and no source divides it between them — so it is drawn as unplaced rather than split. The 3.9 km is DORTS's own figure for the Neihu section, and it is the line's only underground running: the builder records the Muzha section as 全線以高架方式興建, built entirely on viaduct. zh.wikipedia's route description reads 穿越福州山隧道後, which implies a tunnel between BR05 and BR06 where DORTS instead has the line passing the north side of the Xinhai road tunnel. A viaduct does not run through a hill, so the drawing follows the builder — but the two accounts of that half-kilometre have not been reconciled.

Distance is to scale. Height is not: no source gives a rail level, and an axis invented to look like a measurement would be worse than none.



The line as surveyed, from MOTC route geometry — 25.0 km measured along the alignment from the first station to the last. Termini and interchanges are labelled; every station has a name on hover.

Specifications

Route length, operating	25.17 km ^[1]
Route length, published	25.7 km ^[2]
Stations	24 ^[2]
Elevated stations	22 ^[2]
Underground stations	2 ^[2]
Underground running	3.9 km ^[2]
Guideway spacing	1880 mm ^[8]
Electrification	750 V DC ^[8]
Automation grade	GoA4 ^[8]
Peak headway	2–4 min ^[7]
Off-peak headway	4–10 min ^[7]
Late-evening headway	12 min ^[7]
Design minimum headway	80 s ^[8]
Maximum operating speed	70 km/h ^[8]
Design speed	80 km/h ^[8]
End-to-end journey	45 min ^[7]

References

6 of 15 sources on this page are primary — published by the operator, the builder or government.

- Taiwan MOTC TDX — StationOfRoute, operator TRTC** [https://tdx.transportdata.tw/]
PRIMARY Ministry of Transportation and Communications, Taiwan accessed 2026-08-05
archived [https://web.archive.org/web/20260806042022/https://tdx.transportdata.tw/]
Government open data, committed to this repository at data/tdx/. The route's revenue length is the CumulativeDistance against the last station of route BR-1; the RouteLength field is zero for every metro route, which is what once led this page to say the figure was not published.
- Wenshan—Neihu Line** [https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94]
文山內湖線
PRIMARY Taipei City Government, Department of Rapid Transit Systems (DORTS)
accessed 2026-08-06
archived [https://web.archive.org/web/20260209210832/https://www.dorts.gov.taipei/cp.aspx?n=DBAC040496EFAB94]

The builder's own project record for both halves: 木柵線 約10.9公里，設12座車站 and 內湖線 約14.8公里，設12座車站（地下段3.9公里，2座地下站）。Those two lengths are what add to the 25.7 km that circulates as this line's route length.

3 **Linguang station** [<https://zh.wikipedia.org/zh-tw/麟光站>]

麟光站

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-07

The station chronology that dates both 1993 fires and numbers the second one: 「1993年5月5日：辛亥站前發生火燒車事件」 and 「1993年9月24日：發生第二次火燒車事件」。Also the source for this station having been the first completed with two exits after the 1996 opening, and for its two ticket machines being the fewest on the network.

4 **How many trains were procured for each line of the Taipei metro network?** [https://www.dorts.gov.taipei/News_Content.aspx?n=2A66A485FACB0D5B&s=C8602F8588914E91]

台北都會區大眾捷運系統路網中，各路線所採購之列車數為何？

PRIMARY Taipei City Government, Department of Rapid Transit Systems (DORTS)
accessed 2026-08-07

archived [https://web.archive.org/web/20260807044541/https://www.dorts.gov.taipei/News_Content.aspx?n=2A66A485FACB0D5B&s=C8602F8588914E91]

The builder's own procurement table, by contract: 木柵線 CC350 51對電聯車（102輛）and 文湖線 CB370 101對電聯車（202輛）。Cited here for the pair counts that do not divide into the reported train counts.

5 **Taipei Metro rolling stock** [<https://zh.wikipedia.org/zh-hant/臺北捷運列車>]

臺北捷運列車

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-07

archived [<https://web.archive.org/web/20260807045301/https://zh.wikipedia.org/zh-hant/%E8%87%BA%E5%8C%97%E6%8D%B7%E9%81%8B%E5%88%97%E8%BB%8A>]

Source for the 25-and-51 train split, and for the VAL256 running 非固定編組 — four consecutively-numbered cars forming a train rather than a permanent set, which is one of the three readings of the arithmetic problem.

6 **Station architectural design** [<https://www.dorts.gov.taipei/cp.aspx?n=980C85299DA2890A&s=6B0F524CA1EB5C9F>]

車站建築設計

PRIMARY Taipei City Government, Department of Rapid Transit Systems (DORTS)
accessed 2026-08-07

archived [<https://web.archive.org/web/20260807045159/https://www.dorts.gov.taipei/cp.aspx?n=980C85299DA2890A&s=6B0F524CA1EB5C9F>]

The builder's own account of why the stations look as they do, and — from a completely different direction than the project record — a count that identifies the underground pair: 「文湖線內湖段為木柵段之延伸，屬中運量系統，除大直站為地下車站外，其餘共有十座高架車站」。Ten elevated plus Dazhi is eleven of the Neihu section's twelve; the twelfth is Songshan Airport. Also the source for the single-column load constraint on the Muzha section and the 「空中之河」 / 「湖城故事」 framing of the Neihu section.

7 **Routes and headways** [<https://www.metro.taipei/cp.aspx?n=EAD981369A065968>]

路線及班距

PRIMARY Taipei Rapid Transit Corporation accessed 2026-08-06

archived [https://web.archive.org/web/20260803093910/https://www.metro.taipei/cp.aspx?n=EAD981369A065968]

The operator's own published service pattern. Weekday peak about 2–4 min, off-peak about 4–10 min, about 12 min after 23:00, and one-way running time about 45 minutes.

8 **Wenhu line** [https://zh.wikipedia.org/zh-tw/文湖線]

文湖線

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

archived [https://web.archive.org/web/20250714045049/https://zh.wikipedia.org/zh-tw/文湖線]

Carries both length figures in one infobox — 路線長度 25.7公里 against 營運長度 25.18公里 — which is what showed the two were never in conflict. Also the source for the 1,880 mm sleeper-centre spacing, the electrification, the speeds, and the identification of **BR13** and **BR14**.

9 **Muzha Line** [https://zh.wikipedia.org/zh-tw/木柵線]

木柵線

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

archived [https://web.archive.org/web/20250531163800/https://zh.wikipedia.org/zh-tw/木柵線]

The fullest chronology of the original section's construction years — the 1988 Matra contract, the 1993 test fires, the 1996 opening.

10 **Muzha metro line: Matra wins, Taipei to pay NT\$1.64 billion** [https://www.epochtimes.com/b5/5/7/23/n995027.htm]

木柵捷運線 馬特拉勝訴 北市要賠16.4億

SECONDARY 孫友廉 , 陳曉宜 , 鄭學庸 , 自由時報 , 23 July 2005 (via 大紀元 reprint)

accessed 2026-08-06

archived [https://web.archive.org/web/20260807043739/https://www.epochtimes.com/b5/5/7/23/n995027.htm]

The report of the judgment that ended the Matra litigation, cited here for the headline figures; the full account is on the Matra dispute page.

11 **Taipei Metro VAL256 electric multiple unit** [https://zh.wikipedia.org/zh-tw/台北捷運VAL256型電聯車]

台北捷運VAL256型電聯車

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

archived [https://web.archive.org/web/20221006051618/https://zh.wikipedia.org/zh-tw/台北捷運VAL256型電聯車]

12 **Bombardier INNOVIA APM 256 electric multiple unit** [https://zh.wikipedia.org/zh-tw/龐巴迪INNOVIA_APM_256型電聯車]

龐巴迪INNOVIA APM 256型電聯車

SECONDARY 維基百科 (Chinese Wikipedia) accessed 2026-08-06

archived [https://web.archive.org/web/20150531215555/https://zh.wikipedia.org/zh-tw/龐巴迪INNOVIA_APM_256型電聯車]

13 **Safety equipment** [https://www.metro.taipei/cp.aspx?n=c4b618682db27979&s=288E97B4A5E9A725]

安全設備

PRIMARY Taipei Rapid Transit Corporation accessed 2026-08-06

archived [https://web.archive.org/web/20250616230758/https://www.metro.taipei/cp.aspx?n=C4B618682DB27979&s=288E97B4A5E9A725]

14 Lille VAL — project profile [https://www.railway-technology.com/projects/lille_val/]

SECONDARY Railway Technology accessed 2026-08-06

archived [https://web.archive.org/web/20260807044500/https://www.railway-technology.com/projects/lille_val/]

Describes the VAL guideway — running surfaces with lateral H-section guide bars that also carry the 750 V supply. Side guidance, not a central rail.

15 That day, that year: 28 March 1996, Taiwan's first metro line, the Muzha Line, opens [https://www.peoplenews.tw/news/19facc82-2320-47f3-8e14-0f62366535f9]

【那一年的這一天】1996.3.28 台灣第一條捷運「木柵線」通車

SECONDARY 民報 Taiwan People News accessed 2026-08-06

archived [https://web.archive.org/web/20191207073911/https://www.peoplenews.tw/news/19facc82-2320-47f3-8e14-0f62366535f9]

Cited for the "first in Taiwan" claim, which had been on this page as a bare superlative. The live URL rotted some time before 6 August 2026 — the citation survives only because the Internet Archive holds a 2019 capture, which is the archiving policy's whole argument made for it.

Last updated: 2026-08-06

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Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operator TRTC. Government open data. Retrieved 2026-08-05; source last updated 2020-01-31. Provenance.